

Name : _____

HURLSTONE AGRICULTURAL HIGH SCHOOL



TRIAL HIGHER SCHOOL CERTIFICATE

1998

MATHEMATICS 3/4 UNIT (COMMON)

Time Allowed – Two hours
(Plus reading time – 5 minutes)

Examiner – H. Cavanagh.

DIRECTIONS TO CANDIDATES

- This paper contains 8 questions.
- All questions are to be attempted.
- All necessary working should be shown. Marks may not be awarded for careless or badly arranged work.
- Start each question on a new sheet of paper. Ensure that your name is written on each sheet of paper that is submitted.
- Board approved calculators and approved Math-Aids may be used.
- A table of Standard Integrals is supplied for use in the examination.

QUESTION ONE (11 marks) (Start a new sheet of paper)

Ma

- (a) Write $\frac{1 + \sqrt{7}}{3 - \sqrt{7}}$ in the form $a + b\sqrt{7}$ where a and b are rational. 2
- (b) Given $f(x) = \tan^2 x$, find $f'(\frac{\pi}{4})$. 2
- (c) A and B are the points (1, -3) and (6, 7) respectively. P divides the interval AB in the ratio 2 : 3. Find the coordinates of P. 2
- (d) Solve for x : $\frac{6}{x} > x - 1$ 3
- (e) (i) Sketch the graph of $y = |x - 2|$ 2
(ii) For what values of x is $|x - 2| < x$

QUESTION TWO (12 marks) (Start a new sheet of paper)

- (a) Differentiate $e^{4x} \sin x$. 2
- (b) Use the substitution $u = \log_e x$ to evaluate $\int_1^e \frac{(\log_e x)}{x} dx$ 3
- (c) Solve the equation $\sin 2x = \tan x$ for $0 \leq x \leq \pi$. 4
- (d) Evaluate $\int_1^{\sqrt{2}} \frac{dx}{\sqrt{4 - 2x^2}}$ 3

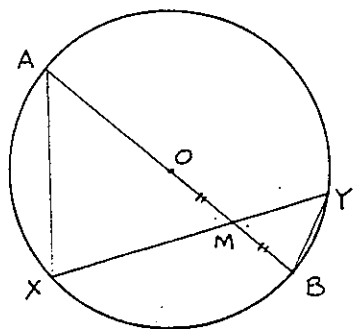
Please turn over to Question Three.....

QUESTION THREE (9 marks) *(Start a new sheet of paper)*

Marks

- (a) (i) Find the domain and range of the function $y = 4\cos^{-1}\frac{x}{3}$ 5
- (ii) Sketch the graph of the function $y = 4\cos^{-1}\frac{x}{3}$ showing clearly the intercepts on the coordinate axes and the coordinates of any endpoints.
- (iii) Find the area of the region in the first quadrant bounded by the curve $y = 4\cos^{-1}\frac{x}{3}$ and the coordinate axes.

- (b) In the diagram below, AB is a diameter of a circle, centre O. The chord XY intersects AB at M, such that OM = BM. 4



- (i) Copy, or trace, the diagram carefully onto your exam paper.
- (ii) Show that $\triangle AXM \cong \triangle YBM$.
- (iii) If $XM = 8$ cms and $YM = 6$ cms, find the radius of the circle.

Please turn over to Question Four.....

QUESTION FOUR (10 marks) *(Start a new sheet of paper)*

Marks

- (a) Using the principles of Mathematical Induction, show that $3^{3n} + 2^{n+2}$ is divisible by 5 for all positive integers n greater than or equal to 1. 5
- (b) Newton's Law of Cooling states that when an object at temperature $T^\circ\text{C}$ is placed in an environment at temperature $T_0^\circ\text{C}$, the rate of temperature loss is given by the equation $\frac{dT}{dt} = k(T - T_0)$ where t is the time in seconds and k is a constant. 5
- (i) Show that $T = T_0 + Ae^{kt}$ is a solution to the equation above.
- (ii) A packet of peas, initially at 24°C is placed in a snap-freeze refrigerator in which the internal temperature is maintained at -40°C . After 5 seconds, the temperature of the peas is 19°C . How long will it take for the peas to reach a temperature of 0°C ?

QUESTION FIVE (12 marks) *(Start a new sheet of paper)*

- (a) (i) Express $2\sin\theta + \cos\theta$ in the form $r\sin(\theta + \alpha)$. 3
- (ii) Hence, or otherwise, solve $2\sin\theta + \cos\theta = 1$ for $0^\circ \leq \theta \leq 360^\circ$.
- (b) $P(2ap, ap^2)$ and $Q(2aq, aq^2)$ are two points on the parabola $x^2 = 4ay$. 9
- (i) If PQ passes through the point $C(2a, 3a)$ show that $pq = p + q - 3$.
- (ii) If M is the midpoint of PQ, show that the coordinates of M are $(a(pq + 3), \frac{a}{2}(pq + 3)^2 - apq)$.
- (iii) Hence show that the locus of the midpoint of all chords passing through $(2a, 3a)$ is $x^2 - 2ax = 2a(y - 3a)$.

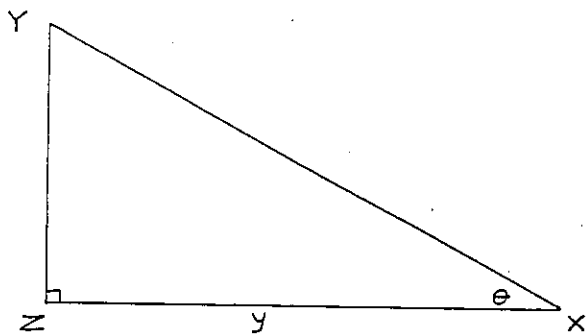
Please turn over to Question Six.....

QUESTION SIX (11 marks) (Start a new sheet of paper)

Marks

- (a) In a group X, there are 7 boys and 3 girls, whilst in another group Y, there are 2 boys and 8 girls. 4
- (i) One person is selected at random from each group. Find the probability that
- (α) both are boys. (β) one is a boy and one is a girl.
- (ii) One group is chosen at random from X and Y and then two people are selected from this group. Find the probability that
- (α) both are boys. (β) one is a boy and one is a girl.

- (b) In $\triangle XYZ$, $ZX = y$, $\angle YZX = 90^\circ$ and $\angle YXZ = \theta$. 7



- (i) Show that the perimeter P of the triangle is given by $P = y(1 + \tan\theta + \sec\theta)$.
- (ii) If $y = 10$ cm and θ is increasing at a constant rate of 0.2 radians / sec find the rate at which the perimeter of the triangle is increasing when $\theta = \frac{\pi}{6}$ radians.

Please turn over to Question Seven

QUESTION SEVEN (10 marks) (Start a new sheet of paper)

Mark

Heavy raindrops are moving horizontally at 36 km/hr in clouds being blown 10 by a steady wind. They then fall 200 metres to the ground below.

(Air resistance may be neglected, and the approximate value $g = 10$ m/sec² may be assumed.)

- (i) Find the time taken for a drop to reach the ground.
- (ii) Find the speed and angle at which a drop hits the horizontal ground.
- (iii) At what angle does a drop hit the ground when the wind speed is doubled?

Please turn over to Question Eight....

QUESTION EIGHT (12 marks) (Start a new sheet of paper)

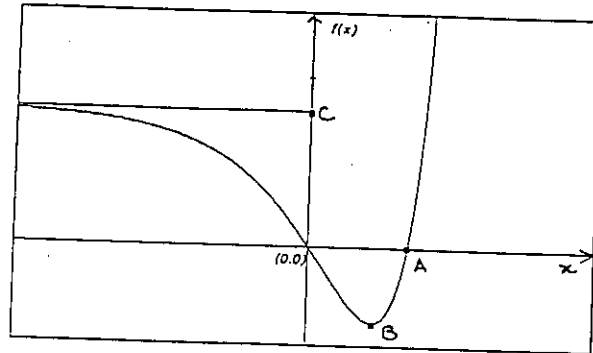
Mark

- (a) The acceleration of a particle, when x metres from the origin on a directed axis, is $(4x - 2x^3) \text{ ms}^{-2}$.

It is released from rest at $x = 2$.

- Determine v^2 as a function of x , where $v \text{ ms}^{-1}$ is the velocity.
- Determine (α) the position at which it next comes to rest,
(β) and its acceleration at that point.

- (b) The graph of $f(x) = e^{2x} - 5e^x + 4$ is shown below



- Write the equation of the horizontal asymptote through C.
- Find the coordinates of A, the point where $y = f(x)$ crosses the x -axis, and B, a stationary point. Leave your answers in exact form.
- Find the equation of the normal through (0,0).
- The normal intersects the curve at another point, N. The x coordinate of N is close to 1.5. Use one application of Newton's Method to find the value of N and round answer correct to three decimal places.

End of Paper.

30 Trial - 1998 - Solutions.

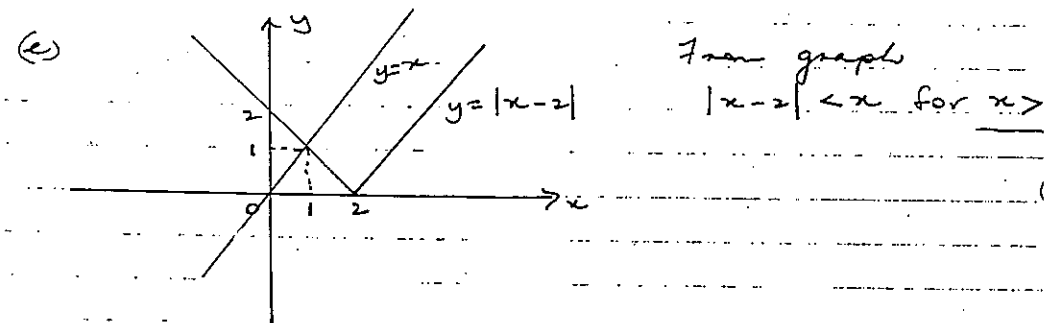
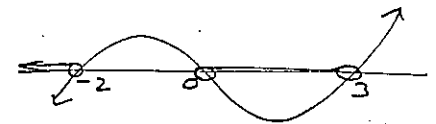
$$\begin{aligned} \textcircled{1} (a) \quad \frac{1+\sqrt{7}}{3-\sqrt{7}} \times \frac{3+\sqrt{7}}{3+\sqrt{7}} &= \frac{3+4\sqrt{7}+7}{9-7} \\ &= \frac{10+4\sqrt{7}}{2} \\ &= 5+2\sqrt{7} \end{aligned}$$

$$\begin{aligned} (b) \quad f(x) &= \tan^2 x \\ f'(x) &= 2 \tan x \cdot \sec^2 x \\ f'\left(\frac{\pi}{4}\right) &= 2 \tan \frac{\pi}{4} \cdot \sec^2 \frac{\pi}{4} \\ &= 2(1)(\sqrt{2})^2 \\ &= 4 \end{aligned}$$

$$\begin{aligned} (c) \quad A(1, -3) \quad B(6, 7) \quad P\left(\frac{2(6)+3(1)}{5}, \frac{2(7)+3(-3)}{5}\right) \\ P(3, 1) \end{aligned}$$

$$(d) \quad \frac{6}{x} > x - 1$$

$$\begin{aligned} (x \neq 0) \quad 6x &> x^3 - x^2 \\ x^3 - x^2 - 6x &< 0 \\ x(x^2 - x - 6) &< 0 \\ x(x-3)(x+2) &< 0 \\ \therefore x &< -2 \text{ or } 0 < x < 3 \end{aligned}$$



$$(2) (a) \frac{d}{dx} e^{4x} \sin x = 4e^{4x} \sin x + e^{4x} \cos x$$

$$= e^{4x} (4 \sin x + \cos x)$$

$$(b) I = \int_1^e \frac{\log_e x}{x} dx$$

let $u = \log_e x$ when $x = e, u = 1$
when $x = 1, u = 0$

$$du = \frac{1}{x} dx$$

$$\therefore I = \int_0^1 u du$$

$$= \left[\frac{u^2}{2} \right]_0^1$$

$$= \frac{1}{2}$$

$$(c) \sin 2x = \tan x \quad 0 \leq x \leq \pi$$

$$2 \sin x \cos x = \frac{\sin x}{\cos x}$$

$$2 \sin x \cos^2 x = \sin x$$

$$\sin x (2 \cos^2 x - 1) = 0$$

$$\therefore \sin x = 0 \text{ or } 2 \cos^2 x - 1 = 0$$

$$x = 0, \pi$$

$$\cos x = \pm \frac{1}{\sqrt{2}}$$

$$x = \frac{\pi}{4}, \frac{3\pi}{4}$$

$$\therefore x = 0, \frac{\pi}{4}, \frac{3\pi}{4}, \pi$$

$$(d) \int_1^{\sqrt{2}} \frac{dx}{\sqrt{4-2x^2}} = \frac{1}{\sqrt{2}} \int_1^{\sqrt{2}} \frac{dx}{\sqrt{2-x^2}}$$

$$= \frac{1}{\sqrt{2}} \left[\sin^{-1} \frac{x}{\sqrt{2}} \right]_1^{\sqrt{2}}$$

$$= \frac{1}{\sqrt{2}} \left\{ \frac{\pi}{2} - \frac{\pi}{4} \right\}$$

$$= \frac{\pi}{4\sqrt{2}}$$

(2)

$$(3) (a) (i) y = 4 \cos^{-1} \frac{x}{3}$$

$$\frac{y}{4} = \cos^{-1} \frac{x}{3}$$

$$\frac{x}{3} = \cos \frac{y}{4}$$

$$x = 3 \cos \frac{y}{4}$$

$$\therefore D: \{-3 \leq x \leq 3\}$$

$$R: \{0 \leq y \leq 4\pi\}$$

$$(iii) A = \int_0^{2\pi} 3 \cos \frac{y}{4} dy$$

$$= \left[12 \sin \frac{y}{4} \right]_0^{2\pi}$$

$$= 12 \left(\sin \frac{\pi}{2} - \sin 0 \right)$$

$$= 12 \text{ units}^2$$

(3)

$$(b) (ii) \text{ In } \Delta's A \times M, Y B M$$

$$\angle A = \angle Y \text{ (angles in same segment, on arc XB)}$$

$$\angle A M X = \angle Y M B \text{ (vert. opp. } \angle's)$$

$$\therefore \Delta A \times M \parallel \Delta Y B M \text{ (equiangular)}$$

(4)

$$(ii) r = 2x$$

$$\therefore \frac{x}{8} = \frac{6}{x+r}$$

$$x(x+r) = 48$$

$$x^2 + x(2x) = 48$$

$$3x^2 = 48$$

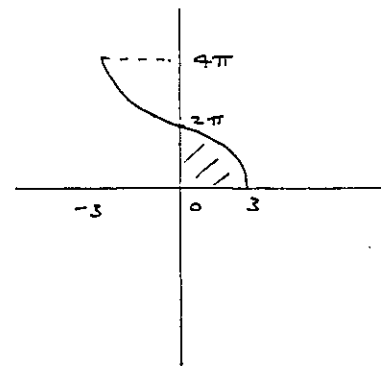
$$x^2 = 16$$

$$x = 4 \text{ (} x > 0 \text{)}$$

$$\therefore r = 8 \text{ cm.}$$

(3)

(ii)



(5)

④ (a) To prove that $3^{3n} + 2^{n+2}$ is divisible by 5 for all $n \geq 1$

Test for $n=1$,

$$3^3 + 2^3 = 27 + 8 \\ = 35$$

True for $n=1$ ($35 = 5 \times 7$)

Assume true for $n=k$

$$\text{i.e. } 3^{3k} + 2^{k+2} = 5M \quad (M \in \mathbb{J})$$

$$\begin{aligned} \text{Consider } 3^{3(k+1)} + 2^{(k+1)+2} &= 3^3 \cdot 3^{3k} + 2^{k+3} \\ &= 3^3(5M - 2^{k+2}) + 2^{k+3} \\ &\quad (\text{since } 3^{3k} = 5M - 2^{k+2}) \\ &= 27 \cdot 5M - 27 \cdot 2^{k+2} + 2^{k+3} \\ &= 27 \cdot 5M - 25 \cdot 2^{k+2} \\ &= 5(27M - 5 \cdot 2^{k+2}) \\ &= 5P \quad P \in \mathbb{J} \end{aligned}$$

Hence, if true for $n=k$, result holds true for $n=k+1$.

Since true for $n=1$, it is true for $n=1+1=2$.

Since true for $n=2$, it is true for $n=2+1=3$.

By principle of mathematical induction, result is true for all positive integral values of n .

④ (b) (i) $\frac{dT}{dt} = k(T - T_0)$

If $T = T_0 + Ae^{kt}$

Then $\frac{dT}{dt} = Ake^{kt}$

$$= k(Ae^{kt})$$

$$= k(T - T_0) \quad \text{Q.E.D.}$$

(ii) When $t=0$, $T=24$, $T_0 = -40$

$$\therefore 24 = -40 + A$$

$$A = 64$$

When $t=5$, $T=19$

$$\therefore 19 = -40 + 64e^{5k}$$

$$59 = 64e^{5k}$$

$$e^{5k} = \frac{59}{64}$$

$$k = \frac{1}{5} \log \frac{59}{64} \quad [k = -0.0162691]$$

For $T=0$

$$0 = -40 + 64e^{kt}$$

$$64e^{kt} = 40$$

$$e^{kt} = \frac{40}{64}$$

$$kt = \log \frac{5}{8}$$

$$t = \frac{1}{k} \log \frac{5}{8}$$

$$= 28.889 \dots$$

$$\therefore \underline{29 \text{ seconds}}$$

$$\begin{aligned}
 \textcircled{5} \textcircled{a} \textcircled{i} \quad 2 \sin \theta + \cos \theta &= \sqrt{5} \left(\frac{2}{\sqrt{5}} \sin \theta + \frac{1}{\sqrt{5}} \cos \theta \right) \\
 &= \sqrt{5} \left(\sin \theta \left(\frac{2}{\sqrt{5}} \right) + \cos \theta \left(\frac{1}{\sqrt{5}} \right) \right) \\
 &= \sqrt{5} \sin(\theta + \alpha) \quad \text{where } \cos \alpha = \frac{2}{\sqrt{5}} \\
 &\quad \sin \alpha = \frac{1}{\sqrt{5}} \\
 &\quad \text{i.e. } \alpha = 26^\circ 34'
 \end{aligned}$$

$$\begin{aligned}
 \textcircled{ii} \quad \therefore 2 \sin \theta + \cos \theta &= 1 \quad \text{for } 0^\circ \leq \theta \leq 360^\circ \\
 \sqrt{5} \sin(\theta + 26^\circ 34') &= 1 \quad \text{for } 26^\circ 34' \leq \theta + 26^\circ 34' \leq 386^\circ 34'
 \end{aligned}$$

$$\sin(\theta + 26^\circ 34') = \frac{1}{\sqrt{5}}$$

S	A
+	-

$$\text{b.a. } \theta + 26^\circ 34' = 26^\circ 34'$$

$$\theta + 26^\circ 34' = 26^\circ 34', 153^\circ 26', 386^\circ 34'$$

$$\therefore \theta = 0^\circ, 126^\circ 52', 360^\circ$$

$$\begin{aligned}
 \text{or } \textcircled{ii} \quad 2 \sin \theta + \cos \theta &= 1 \\
 \text{Using } \sin \theta &= \frac{2t}{1+t^2} \text{ and } \cos \theta = \frac{1-t^2}{1+t^2} \text{ where } t = \tan \frac{\theta}{2}
 \end{aligned}$$

$$\frac{4t}{1+t^2} + \frac{1-t^2}{1+t^2} = 1$$

$$4t + 1 - t^2 = 1 + t^2$$

$$2t^2 - 4t = 0$$

$$2t(t-2) = 0$$

$$t = 0, 2$$

$$\text{i.e. } \tan \frac{\theta}{2} = 0 \quad \tan \frac{\theta}{2} = 2$$

$$\frac{\theta}{2} = 0^\circ, 180^\circ \quad \frac{\theta}{2} = 126^\circ 52'$$

$$\theta = 0^\circ, 360^\circ$$

Also check $\theta = 180^\circ$ is a soln.

$$2(0) - 1 \neq 1 \quad \therefore \theta \neq 180^\circ$$

$$\begin{aligned}
 \textcircled{5} \textcircled{b} \textcircled{i} \quad \text{grad. PQ, } m &= \frac{a(q^2 - p^2)}{2a(q-p)} \\
 &= \frac{q+p}{2}
 \end{aligned}$$

$\therefore$ Eqn. PQ:

$$y - ap^2 = \frac{p+q}{2}(x - 2ap)$$

$$= \frac{1}{2}(p+q)x - ap^2 - apq$$

$$y = \frac{1}{2}(p+q)x - apq$$

If $C(2a, 3a)$ satisfies,

$$3a = \frac{1}{2}(p+q)2a - apq$$

$$3a = ap + aq - apq$$

$$\text{i.e. } pq = p+q-3 \quad \text{Q.E.D.}$$

$$\begin{aligned}
 \textcircled{ii} \quad M &= \left(\frac{2a(p+q)}{2}, \frac{a(p^2+q^2)}{2} \right) \\
 &= (a(p+q), \frac{a}{2}(p^2+q^2))
 \end{aligned}$$

$$\begin{aligned}
 \text{from (i) } pq+3 &= p+q \text{ and } p^2+q^2 = (p+q)^2 - 2pq \\
 &= (pq+3)^2 - 2pq
 \end{aligned}$$

$$\therefore M = \left(a(pq+3), \frac{a}{2}[(pq+3)^2 - 2pq] \right)$$

$$= \left(a(pq+3), \frac{a}{2}(pq+3)^2 - apq \right) \quad \text{Q.E.D.}$$

(iii) Since at M,

$$x = a(pq+3)$$

$$pq+3 = \frac{x}{a}$$

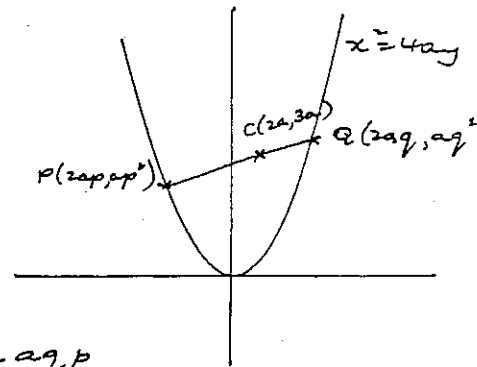
$$pq = \frac{x}{a} - 3$$

$$\text{Then } y = \frac{a}{2} \left(\frac{x}{a} \right)^2 - (x-3a)$$

$$= \frac{x^2}{2a} - x + 3a$$

$$2ay = x^2 - 2ax + 6a^2$$

$$x^2 - 2ax = 2ay - 6a^2$$



⑥ (a) X: 7B, 3G
Y: 2B, 8G

(i) (a) $P(BB) = \frac{7}{10} \times \frac{2}{10} = \frac{7}{50}$

(b) $P(B,G) + P(G,B) = \frac{7}{10} \times \frac{8}{10} + \frac{3}{10} \times \frac{2}{10}$
 $= \frac{56+6}{100} = \frac{31}{50}$

(ii) (a) $P(X, BB) + P(Y, BB) = \frac{1}{2} \times \frac{7}{10} \times \frac{6}{9} + \frac{1}{2} \times \frac{2}{10} \times \frac{1}{9}$
 $= \frac{42+2}{180} = \frac{11}{45}$

(b) $P(X, BG) + P(X, GB) + P(Y, BG) + P(Y, GB)$
 $= \frac{1}{2} \times \frac{7}{10} \times \frac{3}{9} + \frac{1}{2} \times \frac{3}{10} \times \frac{7}{9} + \frac{1}{2} \times \frac{2}{10} \times \frac{8}{9} + \frac{1}{2} \times \frac{8}{10} \times \frac{2}{9}$
 $= \frac{21+21+16+16}{180} = \frac{37}{90}$

(b) (i) $\tan \theta = \frac{y}{x}$
 $\therefore 42 = y \tan \theta$

$\cos \theta = \frac{y}{x \cdot 4}$

$\therefore x \cdot 4 = \frac{y}{\cos \theta} = y \sec \theta$

$P = x^2 + y^2 + x \cdot y$

$= y + y \tan \theta + y \sec \theta$

$= y(1 + \tan \theta + \sec \theta)$ Q.E.D. $= 20$

(ii) $y = 10, \frac{dy}{dt} = 0.2$

$\frac{dP}{dt} = \frac{dP}{dy} \times \frac{dy}{dt}$

$\frac{dP}{dy} = 10(\sec^2 \theta + \sec \theta \tan \theta)$

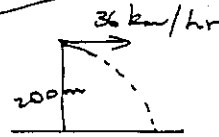
When $\theta = \frac{\pi}{6}$,

$\frac{dP}{dy} = 10\left(\left(\frac{2}{\sqrt{3}}\right)^2 + \frac{2}{\sqrt{3}}\right)$

i.e. $\frac{dP}{dt} = 20 \times 0.2$

$= 4 \text{ cm/sec}$

Q.7



$\frac{dy}{dt} = -10$

$y = -10t + c_1$

$t=0, y=0, \therefore c_1=0$

$y = -10t$

$y = -5t^2 + c_3$

$t=0, y=200, \therefore c_3=200$

$y = -5t^2 + 200$

$\ddot{x} = 0$

$\dot{x} = c_2$

$\dot{x} = 36 \text{ km/hr}$

$= \frac{36 \times 1000}{60 \times 60}$

$\dot{x} = 10 \text{ m/sec}$

$x = 10t + c_4$

$t=0, x=0, c_4=0$

$x = 10t$

(a) (i) Time for $y=0$

$0 = -5t^2 + 200$

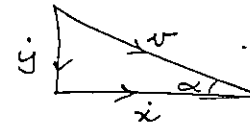
$t^2 = 40$

$t = \pm 2\sqrt{10} \quad (t > 0)$

$\therefore$ drop reaches ground in $2\sqrt{10}$ sec.

(ii) When $t = 2\sqrt{10}$

$\dot{y} = -20\sqrt{10}, \dot{x} = 10$



$\therefore V^2 = (20\sqrt{10})^2 + 10^2$
 $= 4100$

$V = 10\sqrt{41}$

$\approx 64.03 \text{ m/sec}$

$\tan \alpha = \left| \frac{\dot{y}}{\dot{x}} \right|$ for acute α

$= \frac{20\sqrt{10}}{10}$

$= 2\sqrt{10}$

$\alpha \approx 81^\circ$

(ii) Speed = 20 m/sec.

$\therefore \dot{y} = -20\sqrt{10}, \dot{x} = 20$

$\tan \alpha = \sqrt{10}$

$\alpha = 72^\circ 27'$

⑧ (a) $a = (4x - 2x^3) \text{ m/s}^2$
when $t=0, v=0, x=2$.

(i) $a = \frac{d}{dx} \left(\frac{1}{2} v^2 \right) = 4x - 2x^3$

$$\frac{1}{2} v^2 = 2x^2 - \frac{x^4}{2} + C$$

when $x=2, v=0$

i.e. $0 = 8 - 8 + C$

$C = 0$

i.e. $v^2 = 4x^2 - x^4$

(ii) (2) For $v=0, 0 = 4x^2 - x^4$
 $= x^2(4 - x^2)$

i.e. $x = 0, \pm 2$

Since starts at $x=2$, next comes to rest at $x=0$

(3) When $x=0, a=0$

④

⑧ (b) (i) $y = 4$

(ii) at A, $y=0$

i.e. $e^{2x} - 5e^x + 4 = 0$

let $m = e^x$

Then $m^2 - 5m + 4 = 0$

$(m-4)(m-1) = 0$

$m = 4, 1$

$e^x = 4, 1$

$x = \ln 4, \ln 1 (= 0)$

$\therefore$ At A $(\ln 4, 0)$

At B, $\frac{dy}{dx} = 0$

i.e. $2e^{2x} - 5e^x = 0$

$e^x(2e^x - 5) = 0$

$e^x = 0, \frac{5}{2}$

$\therefore$ At B, $x = \ln \frac{5}{2}, y = \frac{25}{4} - 5 \cdot \frac{5}{2} + 4$
 $= -\frac{9}{4}$

i.e. $B(\ln \frac{5}{2}, -\frac{9}{4})$

(iii) At $(0,0) \frac{dy}{dx} = 2 - 5 = -3$

$\therefore$ grad. of normal at $(0,0) = \frac{1}{3}$

Eqn. of normal: $y = \frac{1}{3}x$

$x - 3y = 0$

(iv) At N, $\begin{cases} y = \frac{1}{3}x \\ y = e^{2x} - 5e^x + 4 \end{cases}$

$\therefore \frac{1}{3}x = e^{2x} - 5e^x + 4$

$$x = 3e^{2x} - 15e^x + 12$$
$$3e^{2x} - 15e^x + 12 - x = 0$$

$$\text{let } f(x) = 3e^{2x} - 15e^x + 12 - x.$$
$$f(1.5) = 3e^3 - 15e^{1.5} + 12 - 1.5$$
$$f'(x) = 6e^{2x} - 15e^x - 1$$
$$f'(1.5) = 6e^3 - 15e^{1.5} - 1$$

$$\text{Approx. to } x = 1.5 - \frac{f(1.5)}{f'(1.5)}$$
$$= \underline{1.432}$$